

host

$$\frac{\partial \bar{z}_H}{\partial t} =$$

abiotic selection

toward a local optimum

$$-A_H \bar{z}_H$$

biotic selection

host evades, parasite follows

$$-B_H (\bar{z}_P - \bar{z}_H)$$

dispersal

offspring settle nearby

$$+ \frac{\sigma_H^2}{2} \nabla^2 \bar{z}_H$$

random genetic drift

space-time white noise

$$+ D_H \xi_H$$

parasite

$$\frac{\partial \bar{z}_P}{\partial t} =$$

$$-A_P \bar{z}_P$$

$$+ B_P (\bar{z}_H - \bar{z}_P)$$

$$+ \frac{\sigma_P^2}{2} \nabla^2 \bar{z}_P$$

$$+ D_P \xi_P$$



Its solutions:

two random surfaces

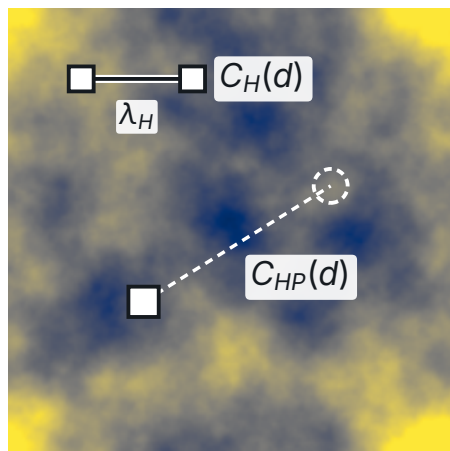
a bivariate Gaussian random field

each with its own spatial scale

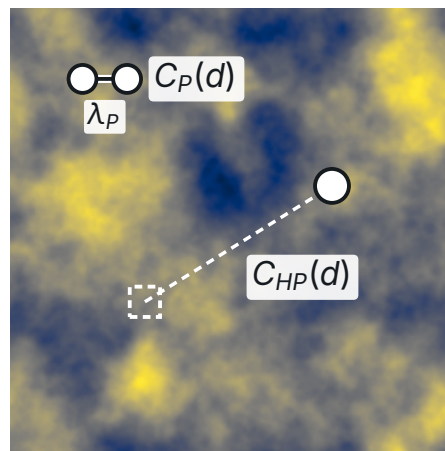
$$\lambda_H = \sigma_H / \sqrt{A_H}$$

$$\lambda_P = \sigma_P / \sqrt{A_P}$$

host mean trait



parasite mean trait



$C_H(d)$, $C_P(d)$
intraspecific spatial
autocovariance functions

$C_{HP}(d)$
interspecific spatial
cross-covariance function

dashed outline: the other species